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TWiki > NTOF Web > Lxplus (2026-05-07, RobertoZarrella)

-- [FrankGunsing](#) - 2015-05-04

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1 Be sure to be part of the computer group 'za'.

In order to use lxplus and access n\_TOF data, you have to belong to the computer group 'za'. Check [here](#) if this is the case. If not, self-subscribe to the e-group 'za' at [this page](#).

2 Setting up a minimal lxplus environment (updated 2024-06-04)

Data analysis at n\_TOF is usually done using the linux cluster (**lxplus**). If you are not familiar with the command line in linux, read up on this topic first. You can login to lxplus from any computer using your cern account by using ssh on your local computer

```
ssh -XY yourcernlogin@lxplus.cern.ch
Password:
```

and enter your CERN password (**CERN Single Sign On (SSO)**). Now you are connected to lxplus and you will get the shell prompt from where you can issue commands. The shell you are in is most probably bash. The shell can be changed by [managing your account from here](#). Find out which shell you have by typing:

```
echo $SHELL
```

A file needs to be setup to define your environment. In the case of a bash shell this is ~/.bash\_profile (note the dot). You can edit this file using a text editor, for example nano.

```
nano ~/.bash_profile
```

This file is executed at every login, and is the best place to set environmental variables and other settings. The standard environment, without a .bash\_profile, already provides access to ROOT and other programs. You can execute a custom environment in order to have more programs and usually more recent versions, maintained by [CERN-SFT](#) or commands to facilitate access to CTA and [EOS](#). A customised .bash\_profile can be for example:

```
#!/bin/bash

# display message
echo "executing bash_profile"

# customized prompt colors
bldgrn='\[\e[1;32m\' # bold green
bldred='\[\e[1;31m\' # bold red
txtblk='\[\e[0;30m\' # black
export PS1="${bldgrn}\A ${bldred}\u@H:\W > ${txtblk}"

# access to pre-compiled n_TOF executable scripts and codes (including SAMMY), sources in /eos/experiment/ntof/src
export PATH=$PATH:/eos/experiment/ntof/bin

# more error-safe EOS access
export EOS_MGM_URL=root://eospublic.cern.ch
```

In your home directory, you can define symbolic links to directories often needed:

```
ln -s /eos/experiment/ntof/ ~
```

If you want to set a custom environment with pre-available versions of other programs (ROOT, GEANT4, gcc, python, etc...), you can use one of the LCG distributions (see full list at at <https://lcginfo.cern.ch>) by adding in your .bash\_profile

```
source /cvmfs/sft.cern.ch/lcg/views/setupViews.sh LCG_105 x86_64-el9-gcc13-opt
```

Keep in mind that after this command is issued, you will not be able to use the executables in /eos/experiment/ntof/bin.

3 Large scale data processing (updated 2025-03-13)

To process a set of runs from a specific campaign, one can login to LXPLUS and use the dedicated script available in the /eos/experiment/ntof/repositories/processingscripts. The script is called RunProcessing.sh and the available options are:

```
/eos/experiment/ntof/repositories/processingscripts/RunProcessing.sh
-y Year of the experimental campaign (e.g. 2021)
-a Area (e.g. EAR1/EAR2/EAR3/LAB, ear8/ear1/ear2/lab before 2018, None before 2014)
-c Campaign name (i.e. the "Exp. / Archive Folder" field of the DAQ)
-r First run number
-l Last run number [optional, default = first run number]
-p PSA UserInput file path
-o Output folder [optional, default = current folder]
-s Skip already processed runs [optional, default = 0]
```

This list of options can also be displayed by calling /eos/experiment/ntof/repositories/processingscripts/RunProcessing.sh -help.

The script looks for data in either [EOS](#) or on CTA (tape). If the data are stored on tape, the script stages the files and checks when they are available. As soon as the files are online, the processing starts automatically. **There is no need to manually stage the files and/or launch a dedicated script nor to add any command to your acrontab.**

A couple of important operational notes are provided below:

- The script must be run from a path inside /afs/. For each of the processed runs, a folder containing auxiliary files will be created inside the current directory (i.e. from where you launch the script). **These files will not be deleted automatically and the deletion is left to the user.**
- It is mandatory to indicate the first run to process (-r), while the last run (-l) is not. If no last run is provided, only the first run will be processed.

- The PSA [UserInput](#) path can be provided either as relative or full path. However, if Pulse Shape Fitting is activated, the files containing the pulse shapes must be referenced by their full path inside the [UserInput](#). **Please, make sure to have a fully functioning [UserInput](#) before launching the processing.**
- The output folder can be set either in [EOS](#) or in AFS. By default the output folder is set to the current working directory. It is not possible to set the output folder to `/afs/cern.ch/user` because of the limited storage in this path.
- If the chosen output folder does not exist, the script can create it for the user.
- Output ROOTfiles are saved inside the output folder in two dedicated subfolders:
  - `completed/` which will contain the partial ROOTfiles. These can be deleted if all jobs are completed with no error, but **this cleanup operation is left to the user.**
  - `done` which will contain the final ROOTfiles, named `runXXXXXX.root`
- The `-s` option gives the possibility to only process the runs that have not been processed before. The script looks for ROOTfiles in the `completed` and `done` folders and skips all runs which already have produced ROOTfiles. This is normally not needed, but can be activated by adding `-s 1` to the command.

An example of how to run the script is:

```
/eos/experiment/ntof/repositories/processingscripts/RunProcessing.sh -y 2022 -a EAR1 -c Mo_campaign -o /eos/user/y/yourname/outputPath -p /full/path/to/PSA_UserInput.txt -r 114180 -l 114190
```

3.1 Raw data location and file staging (updated 2025-5-19)

Raw data are stored in two paths simultaneously:

- On [EOS](#), data can be found at `/eos/experiment/ntof/DAQ`. These files can be explored through the normal `ls` command. **Here, data are stored only momentarily, i.e. for two weeks from the moment they were acquired.** After this time, data files can only be found on CTA.
- On CTA (tape), data can be found at `/eos/ctapublicdisk/archive/ntof/`. One can explore these files through the `xrdfs` command in the following way:

```
xrdfs root://eosctapublicdisk.cern.ch/ ls /eos/ctapublicdisk/archive/ntof/
```

Raw data files can be downloaded to tune the PSA parameters before large-scale processing. If the file you want to download is still on EOS, it is sufficient to copy it using a normal `cp` command. If the file is only on CTA, you might need to stage it before you can download it. To check the status of a file, you can launch the command:

```
xrdfs root://eosctapublicdisk.cern.ch/ query prepare 0 /eos/ctapublicdisk/archive/ntof/<path_to_file>
```

This command will show a prompt like the following one:

```
{
  "request_id" : "0",
  "responses" :
  [
    {
      "error_text" : "",
      "has_reqid" : false,
      "on_tape" : true,
      "online" : false,
      "path" : "/eos/ctapublicdisk/archive/ntof/<path_to_file>",
      "path_exists" : true,
      "req_time" : "",
      "requested" : false
    }
  ]
}
```

The reported example shows that the file is on tape (`"on_tape" : true`) but not online (`"online" : false`), i.e. it is not currently accessible. To request the file, issue the command:

```
xrdfs root://eosctapublicdisk.cern.ch/ prepare -s -f /eos/ctapublicdisk/archive/ntof/<path_to_file>
```

From now on, the file will show as requested (`"requested" : true`) and it will become available in up to 72 hours (usually some hours are enough). As soon as the file is online, it can be downloaded using:

```
xrdcp root://eosctapublicdisk.cern.ch/eos/ctapublicdisk/archive/ntof/<path_to_file> /your/path/on/lxplus
```

If you want to stage the files of an entire run, a dedicated script is available `/eos/experiment/ntof/repositories/processingscripts/StageRuns.sh`. The available options are:

```
/eos/experiment/ntof/repositories/processingscripts/StageRuns.sh
-y      Year of the experimental campaign (e.g. 2021)
-a      Area (e.g. EAR1/EAR2/EAR3/LAB, ear0/ear1/ear2/lab before 2018, None before 2014)
-c      Campaign name (i.e. the "Exp. / Archive Folder" field of the DAQ)
-r      First run number
-l      Last run number [optional, default = first run number]
```

The options are the same used by the `RunProcessing.sh` script. The list of options can also be displayed through the command `/eos/experiment/ntof/repositories/processingscripts/StageRuns.sh -help`.

The script also submits a series of condor jobs that continuously check the status of the staged files. These jobs are shown in the `condor_q` print-out, grouped under a `BATCH_NAME` called "stage.dag". All the files of the run are staged when the jobs are completed.

4 Commonly used codes (updated 2018-2-19)

Computer codes of interest for the n\_TOF Collaboration can be made available by putting them in a shared environment. If you decide to share your code of common interest through the n\_TOF workspace on EOS, please do the following:

- put anything executable (binary or scripts) in `/eos/experiment/ntof/bin`
- put source code (or a link to it) in `/eos/experiment/ntof/src`, use a separate directory for each code
- do not compile or write in these directories if you are not the maintainer/contact, use your personal workspace for that
- give a short description of the executable in the table below:

| Codes for n_TOF |                                  |         |                                                                                                                                       |
|-----------------|----------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| executable      | directory name for source        | contact | short description                                                                                                                     |
| RF2sammy        | RF2sammy                         | FG      | Converts TH2 resolution function in SAMMY or REFIT to the numerical UDR format. Type <code>RF2sammy</code> for a short description    |
|                 | TTOFSort                         | FG      | collection of ROOT routines for time-of-flight data processing, see <a href="#">documentation</a> .                                   |
|                 | TWF                              | FG      | ROOT routines for calculating the weighting function from simulated detector responses, see file <code>TWF.h</code> for documentation |
| sort_runs       | sort_runs                        | FG      | sort a run by <code>BunchNumber</code> which is not the case by default for many-channel detectors split over different racks         |
| sammy           | <a href="#">more information</a> | FG      | sammy executables, for testing purposes only. <a href="#">Get your official copy.</a>                                                 |
|                 |                                  |         |                                                                                                                                       |

<https://twiki.cern.ch/twiki/bin/view/NTOF/RYA>

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